Lecture 10. Simulation

Inverse transform, rejection method, discrete distributions, variance reduction

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10.1 Random numbers and simulation

When a probability or expectation is too complicated to compute, we can estimate it by **simulation**: generate many independent realizations of the experiment and use the strong law of large numbers. Everything is built from **random numbers**, independent $U(0,1)$ variables (in practice produced by deterministic pseudorandom generators). For instance, $\lfloor nU \rfloor + 1$ is uniform on $\{1, \dots, n\}$.

Example 10.1 (A random permutation)

To generate a uniformly random permutation of $1, \dots, n$: start from any ordering $P_1, \dots, P_n$; for $k = n, n-1, \dots, 2$, choose I uniform on $\{1, \dots, k\}$ and swap P_I and P_k. Position n receives each element with probability $1/n$, then position $n-1$ each remaining element with probability $1/(n-1)$, and so on, so every permutation has probability $1/n!$. The first r positions filled give a random r-subset, which is how subjects are randomly assigned to treatments.

10.2 Simulating continuous random variables

Motivation. Computers produce uniform random numbers; every other distribution must be built from them. The inverse transform and rejection methods are the two general techniques.

10.2.1 The inverse transformation method

Proposition 10.1

If F is a continuous cdf and $U \sim U(0,1)$, then $X = F^{-1}(U)$ has cdf F.

Proof. $P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$, since F is increasing. ∎

Example 10.2

(a) Exponential: $F(x) = 1 - e^{-\lambda x}$ gives $X = -\frac{1}{\lambda}\log(1-U)$, or equivalently $-\frac{1}{\lambda}\log U$, since $1-U$ is also uniform.
(b) Gamma(n, λ) for integer n: add n independent exponentials, $X = -\frac{1}{\lambda}\log(U_1 U_2 \cdots U_n)$, which needs only one logarithm.

10.2.2 The rejection method

Suppose we can simulate from a density g, and $f(y)/g(y) \le c$ for all y.

Theorem 10.2 (Rejection method)

Repeat: generate $Y \sim g$ and an independent $U \sim U(0,1)$, until $U \le f(Y)/\{cg(Y)\}$; then set $X = Y$. The resulting X has density f, and the number of iterations is geometric with mean c.

Proof. In one iteration, $P(\text{accept}) = \int P(U \le \frac{f(y)}{cg(y)}) g(y)\, dy = \int \frac{f(y)}{c}\, dy = \frac{1}{c}$, and

$$P(Y \le x \mid \text{accept}) = c \int_{-\infty}^{x} \frac{f(y)}{cg(y)} g(y)\, dy = F(x).$$

Iterations are independent, so the number needed is geometric with success probability $1/c$, and the accepted value has the conditional law just computed. ■

Example 10.3 (Normal from exponentials)

$|Z|$ has density $f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$, $x > 0$. With $g(x) = e^{-x}$,

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}}\, e^{x - x^2/2} \le \sqrt{\frac{2e}{\pi}} = c \approx 1.32,$$

the maximum being at $x = 1$, and $\frac{f(x)}{cg(x)} = \exp\{-(x-1)^2/2\}$. So: generate $Y = -\log U_1$, accept if $U_2 \le e^{-(Y-1)^2/2}$ (equivalently if an independent exponential exceeds $(Y-1)^2/2$), and finally attach a random sign. About 1.32 iterations are needed per normal.

Example 10.4 (The polar method)

By Example 6.14, if $R^2 \sim \text{Exp}(1/2)$ and $\Theta \sim U(0, 2\pi)$ are independent, then $R\cos\Theta$, $R\sin\Theta$ are independent standard normals. To avoid computing sines and cosines, generate $V_i = 2U_i - 1$ $(i = 1, 2)$ until $S = V_1^2 + V_2^2 \le 1$; then (V_1, V_2) is uniform on the unit disc, S is uniform and independent of the angle, and $\cos\Theta = V_1/\sqrt{S}$, $\sin\Theta = V_2/\sqrt{S}$. With $R^2 = -2\log S$,

$$X = V_1\sqrt{\frac{-2\log S}{S}}, \qquad Y = V_2\sqrt{\frac{-2\log S}{S}}$$

are independent standard normals. The acceptance probability is $\pi/4$.

A χ^2_n variable can be simulated as $\sum Z_i^2$, or for even n as $-2\log(U_1 \cdots U_{n/2})$ (a Gamma$(n/2, 1/2)$), adding one Z^2 for odd n.

10.3 Simulating from discrete distributions

For $P(X = x_j) = p_j$, set $X = x_j$ if $\sum_{i<j} p_i \le U < \sum_{i \le j} p_i$; this happens with probability p_j. It is the discrete inverse transform.

Example 10.5

(a) Geometric: $X = j$ iff $1 - q^{j-1} \le U < 1 - q^j$ $(q = 1 - p)$, i.e. $X = 1 + \lfloor \log U / \log q \rfloor$ (using $1 - U$ in place of U).
(b) Binomial: $X = \sum_{i=1}^n \mathbb{1}\{U_i < p\}$.
(c) Poisson(λ): with $N(1)$ the number of events of a rate-λ process in $[0, 1]$, and interarrival times $-\log U_i/\lambda$,

$$N = \max\Big\{n : \prod_{i=1}^{n} U_i \ge e^{-\lambda}\Big\}$$

is Poisson(λ): multiply random numbers until the product drops below $e^{-\lambda}$, and subtract 1 from the count.

10.4 Variance reduction

To estimate $\theta = E[X]$ we average n independent copies, with error of order $\sqrt{\mathrm{Var}(X)/n}$. Reducing the variance of what is averaged is as good as increasing n.

10.4.1 Antithetic variables

If X_1, X_2 have the same distribution, $\mathrm{Var}(\frac{X_1+X_2}{2}) = \frac{1}{4}\{2\,\mathrm{Var}(X_1) + 2\,\mathrm{Cov}(X_1, X_2)\}$, which is smaller than for independent copies when $\mathrm{Cov}(X_1, X_2) < 0$. If $X_1 = h(U_1, \dots, U_m)$ with h monotone in each argument, then $X_2 = h(1 - U_1, \dots, 1 - U_m)$ has the same distribution and (as can be shown) $\mathrm{Cov}(X_1, X_2) \le 0$. So each set of random numbers is used twice, once "reversed".

10.4.2 Variance reduction by conditioning

By the conditional variance formula (Proposition 7.6), $\mathrm{Var}(E[X \mid Y]) \le \mathrm{Var}(X)$, and $E[X \mid Y]$ has the same mean θ. So, if $E[X \mid Y]$ can be computed, simulate Y and average $E[X \mid Y]$ instead of X.

Example 10.6 (Estimating π)

With V_1, V_2 independent $U(-1, 1)$ and $I = \mathbb{1}\{V_1^2 + V_2^2 \le 1\}$, $E[I] = \pi/4$ and $\mathrm{Var}(I) = \frac{\pi}{4}(1 - \frac{\pi}{4}) \approx 0.1685$. Conditioning on V_1,

$$E[I \mid V_1] = P(V_2^2 \le 1 - V_1^2 \mid V_1) = \sqrt{1 - V_1^2},$$

so we may average $\sqrt{1 - U^2}$ over random numbers U (the sign of V_1 is irrelevant): $\mathrm{Var}(\sqrt{1 - U^2}) = \frac{2}{3} - \frac{\pi^2}{16} \approx 0.0498$, a 3.4-fold reduction. Since $\sqrt{1 - u^2}$ is monotone, antithetic pairs $\frac{1}{2}\{\sqrt{1 - U^2} + \sqrt{1 - (1 - U)^2}\}$ do better still: their variance is about 0.0069 per pair, against 0.0249 for two independent values.

10.4.3 Control variates

Suppose Y is simulated alongside X and $\mu_Y = E[Y]$ is known. Then $X + c(Y - \mu_Y)$ also has mean θ, and its variance $\mathrm{Var}(X) + c^2\,\mathrm{Var}(Y) + 2c\,\mathrm{Cov}(X, Y)$ is minimized by

$$c^* = -\frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(Y)}, \qquad \mathrm{Var}(X + c^*(Y - \mu_Y)) = \mathrm{Var}(X)(1 - \rho(X, Y)^2).$$

The more strongly Y is correlated with X, the larger the gain; in practice c^* is estimated from the simulated pairs. In Example 10.6, $Y = U^2$ with $E[Y] = 1/3$ is a natural control variate for $\sqrt{1 - U^2}$.